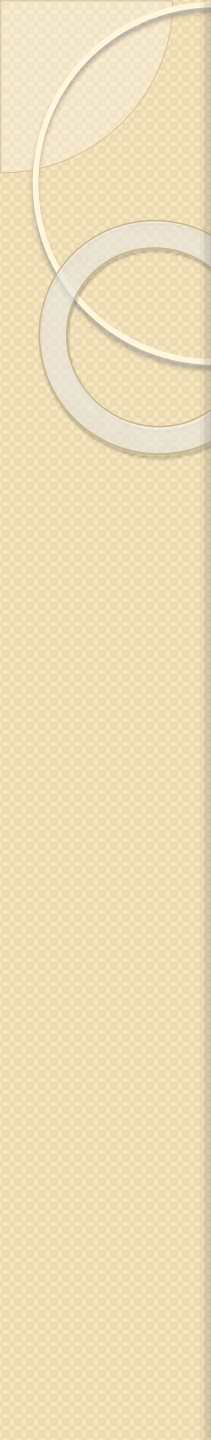


FILE MANAGEMENT in OS

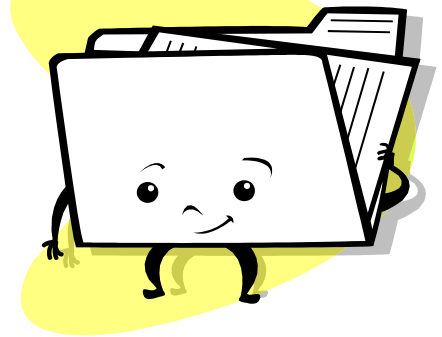


Contents

- File Concepts
- File Types
- File Operations
- Access Methods
- Directory Structure
- File System Structure

File Concept

- The file is a collection of related information that is recorded on storage device.
- The files stores information and data.
- It is a way of data collection which is used as medium of giving input and collecting output from any application.



File Types

1. Ordinary Files (Regular Files)

- It contains user information
- It contains text or executable programs
- It is either text file or binary file
- User can apply various operations

2. Directory Files

- It is like a folder that contains files or other folder
- It don't contain data
- But they have references to file contain within them

File Types

3. Device File

- It contains device information
- It is used to communicate with hardware
- Device file is of two types
 - i. **Character special file**

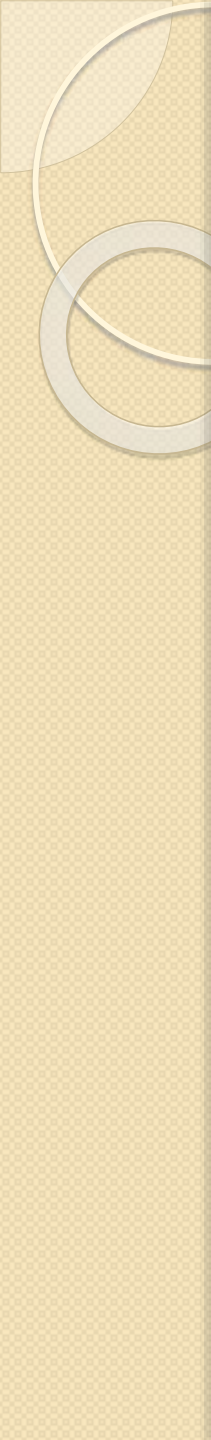
It is a hardware file which read or write data character by character
e.g.: mouse, printer
 - ii. **Block file**

It contains data related to hard disk
e.g.: file related to hard disk

File Types

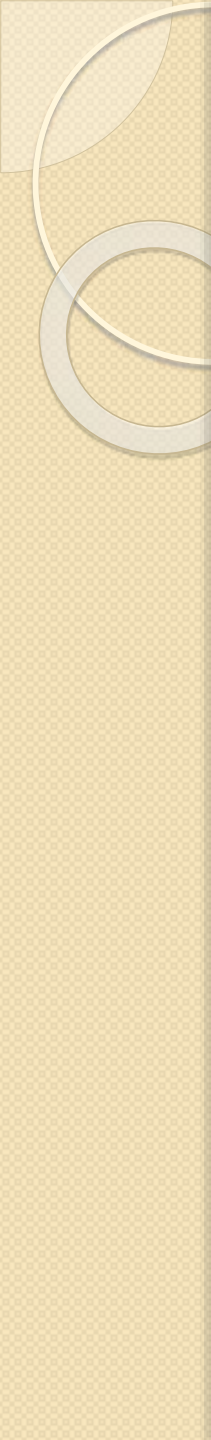
4. FIFO Files

- It is a file which provides temporary buffer for two or more process to communicates by writing data & reading data from buffer
- The buffer associated with this file is allocated when any process opens the file & buffer is discarded when all processes which are connected to this file close their reference



File Operations

1. Create
2. Delete
3. Open
4. Close
5. Read
6. Write
7. Append
8. Seek
9. Get Attributes
10. Set Attributes
11. Rename



Directory Structure

- To keep track of files, file systems normally have directories of folders.
 - Directories are system files for maintaining the structure of the file system.
1. Single Level Directory Structure
 2. Hierarchical Directory
 - i. Two Level Directory Structure
 - ii. Tree Level Directory Structure
 - iii. Acyclic Graph Directory

Single Level Directory

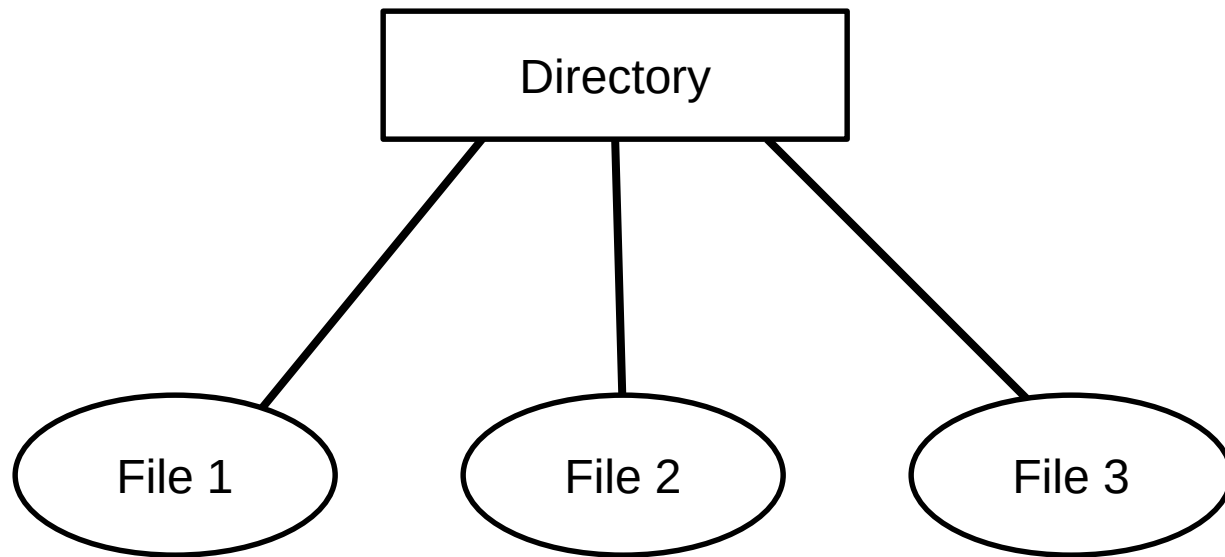


Figure : Single Level Directory

Hierarchical Directory

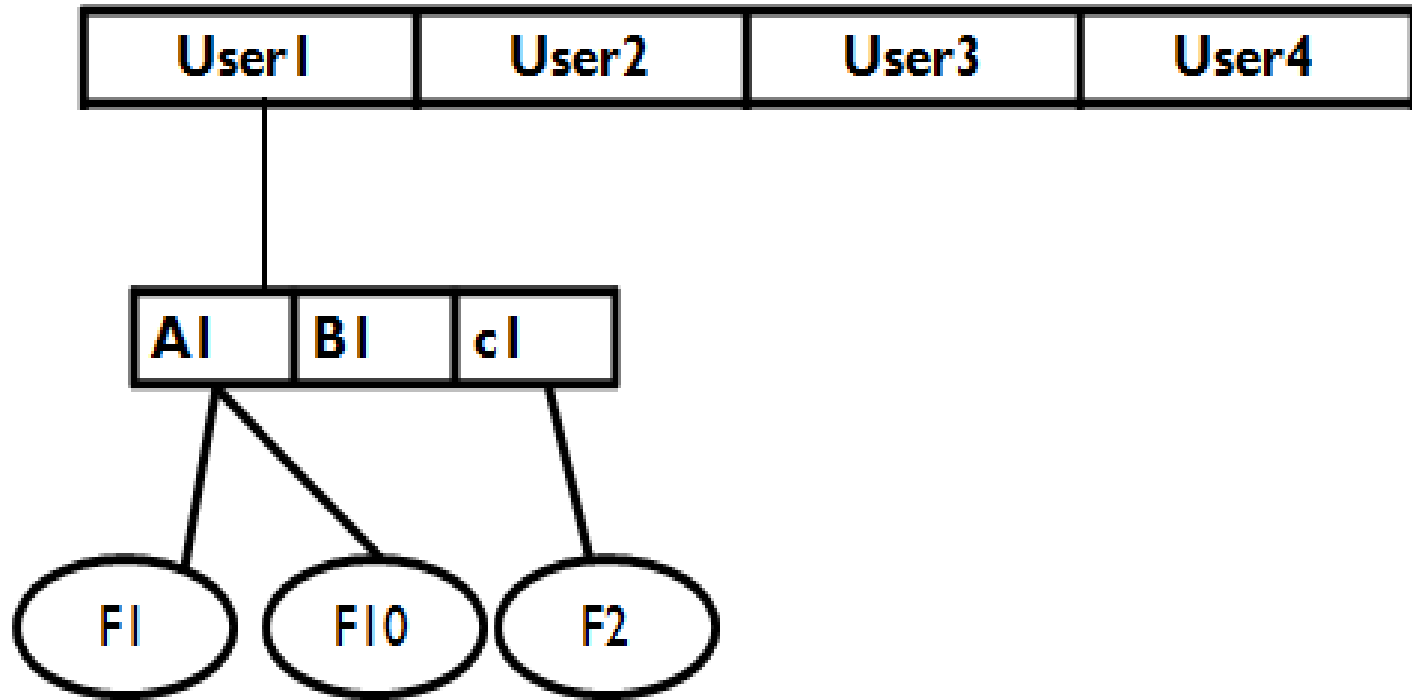


Figure : Two Level Directory

Hierarchical Directory

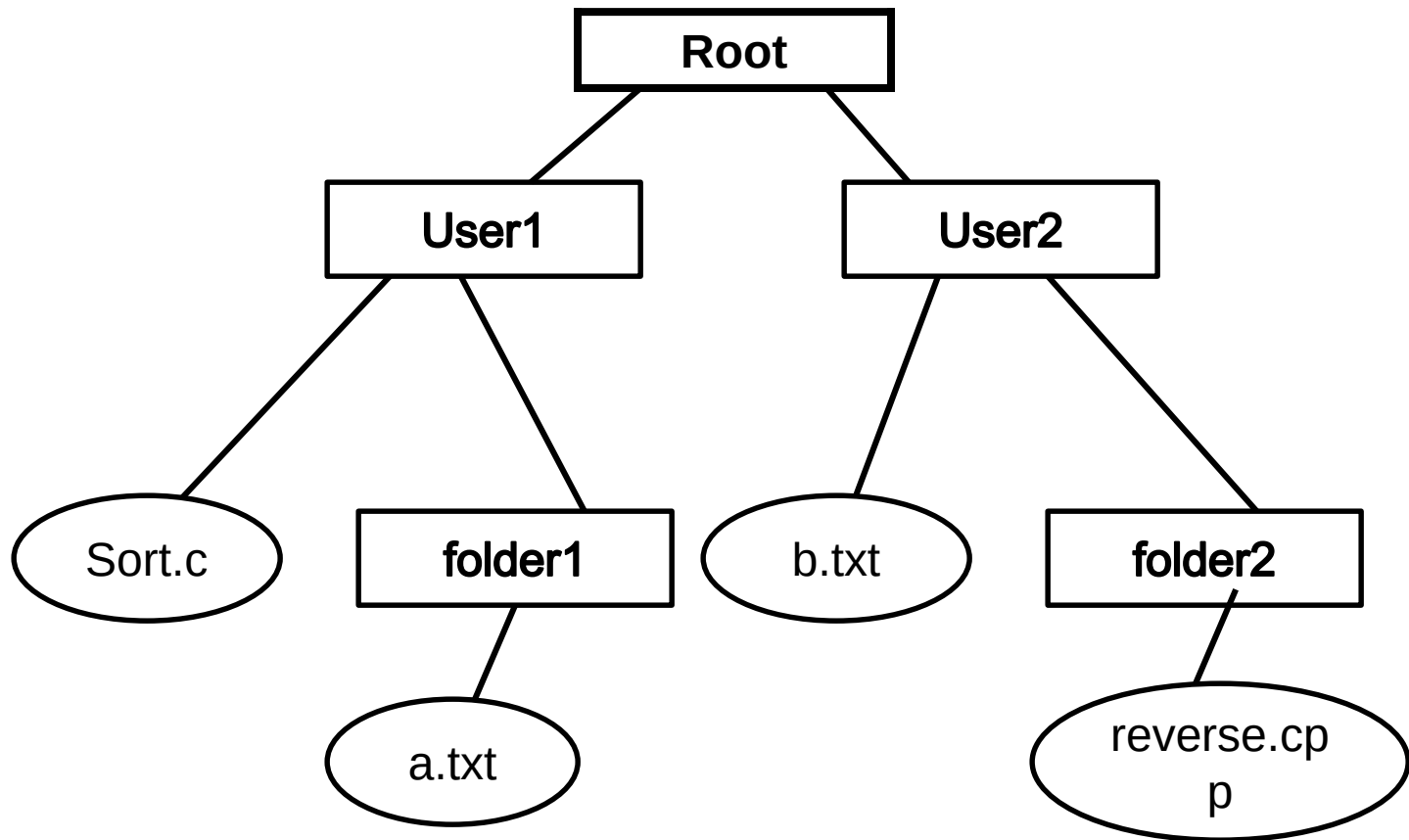


Figure : Tree Level Directory

Hierarchical Directory

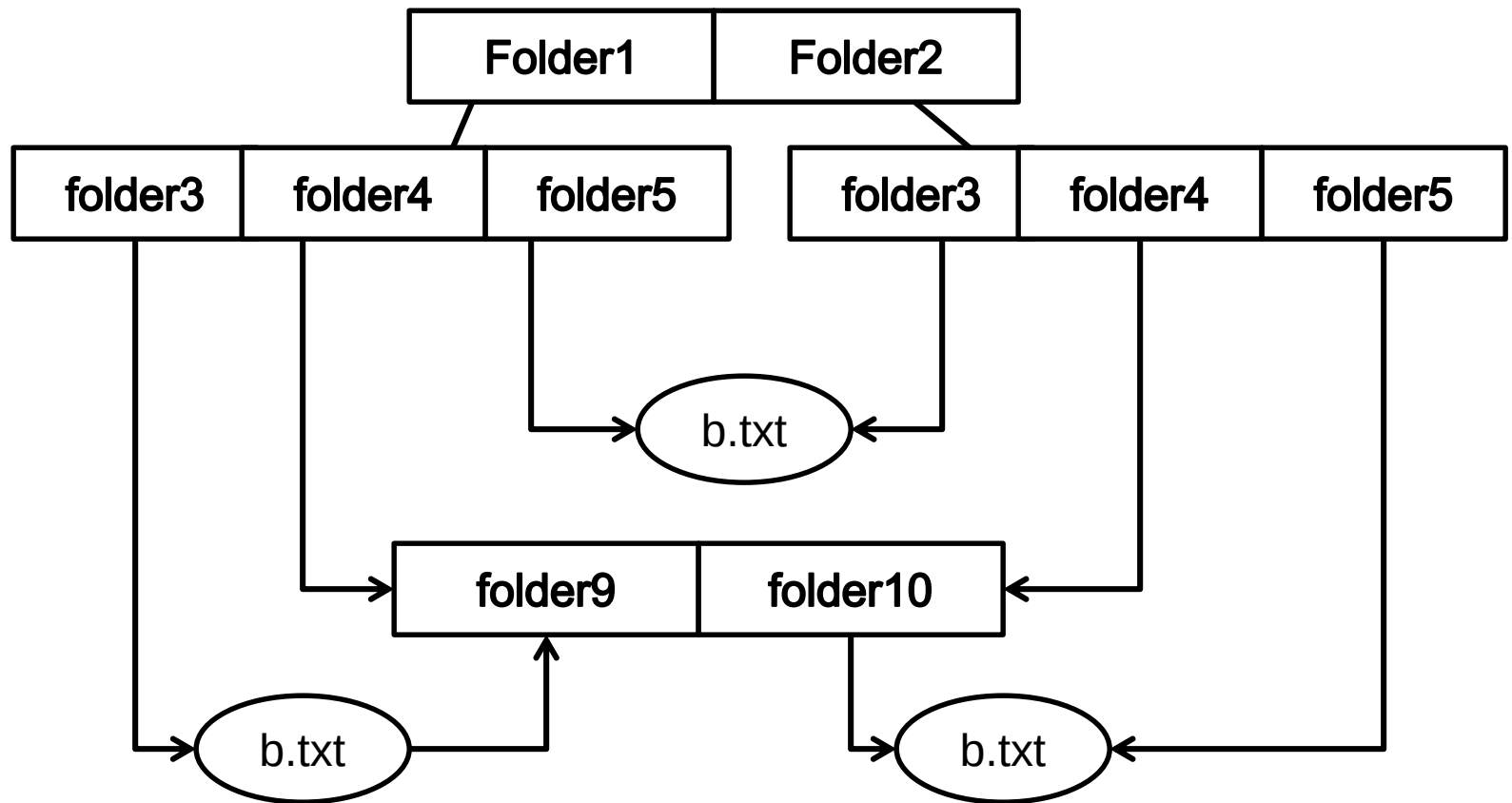
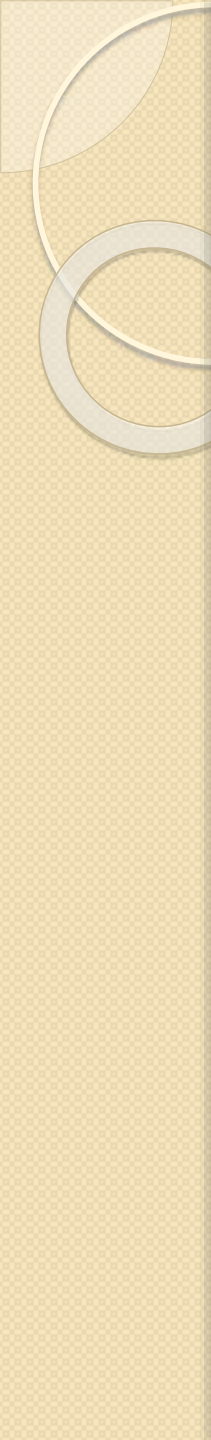


Figure : Acyclic Directory

File System Structure

- The concept of file system is to access the disk efficiently. It allows data to store & retrieve from disk.
- File system structure is composed of many layers as following:
 1. **Input-Output Control Interface**
 - ✓ It consist of device driver and interrupt handler.
 - ✓ Both are used for data transfer between memory & disk
 - ✓ The device driver translates file level command & output is hardware instruction



2. Basic File System Layers

- ✓ It generates commands for device driver
- ✓ Device driver read or write physical block on disk

3. File Organization Module Layer

- ✓ This layer translates logical block address to physical block address

4. Logical System Layer

- ✓ It manages metadata of any file like location & permission of file

5. Application Program Layer

- ✓ This is a layer in which user creates an application program

Access Methods

- The information stored in the file needs to be accessed and read into the computer memory.
- There are different methods available to do it.
 - 1) Sequential Access
 - 2) Indexed Access
 - 3) Direct Access

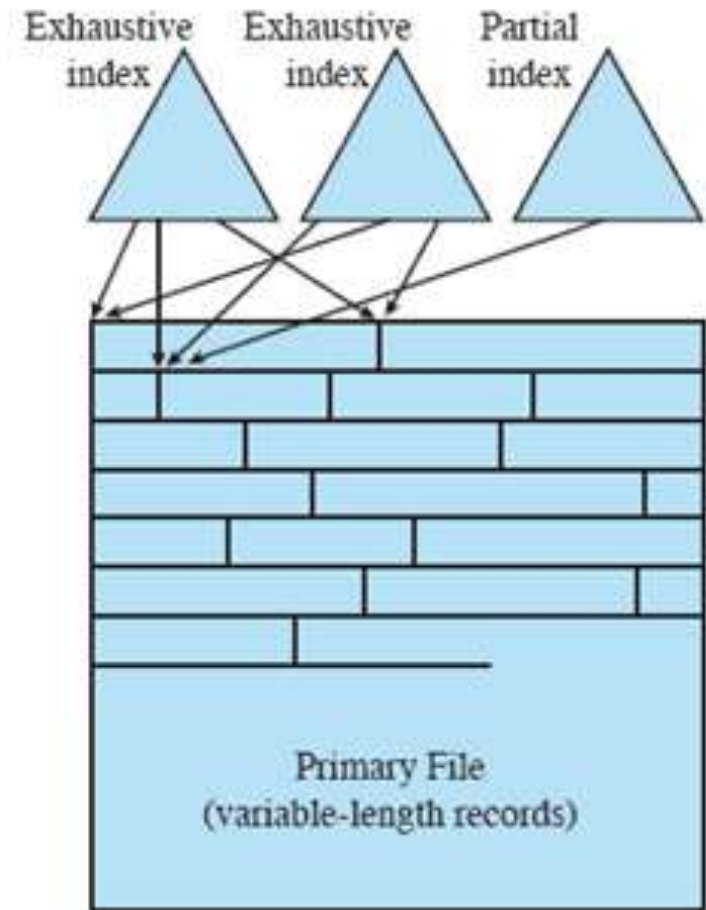
Sequential Access

- Most common form of file structure
- A fixed format is used for records
- Key field uniquely identifies the record & determines storage order
- Typically used in batch applications
- Only organization that is easily stored on tape as well as disk

Fixed-length records
Fixed set of fields in fixed order
Sequential order based on key field

Indexed Access

- Records are accessed only through their indexes
- Variable-length records can be employed
- Exhaustive index contains one entry for every record in the main file
- Partial index contains entries to records where the field of interest exists
- Used mostly in applications where timeliness of information is critical
- Examples would be airline reservation systems and inventory control systems



Direct Access

- Access directly any block of a known address
- Makes use of hashing on the key value
- Often used where:
 - very rapid access is required
 - fixed-length records are used
 - records are always accessed one at a time

[illegible]

References

- Operating System By Andrew Tanenbaums

THANK YOU